

Dijkstra's Algorithm

Input: a graph $G=(V,E)$, nonnegative weight $w: E \rightarrow \mathbb{N}$, source node $s \in V$.

Output: $d: V \rightarrow \mathbb{N}$ where $d(v) = \text{dist}(s,v)$.

Let $n = |V|$, $m = |E|$. Dijkstra runs in $O(m + n \log n)$ time.

Maintain a set $S \subseteq V$ of visited nodes; If $v \in V$ is visited, then $d(v)$ is shortest distance.

On input $G=(V,E)$, $w: E \rightarrow \mathbb{N}$, $s \in V$,

~~Let $S = \{s\}$, $d(s) = 0$~~ Let $d(s) = 0$ Insert $(v, w(s,v))$ into PQ for each $v \neq s$.

~~while $S \neq V$~~ ^{while} PQ can extract some $(v, d(v))$ with min priority

~~choose $u \in V - S$ st $d_u = \min_{u \in S} d(u) + w(u,v)$ is the least~~

Let $d(v) = d_u$

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for each u st. $(v,u) \in E$

~~add v to S~~

decrease priority of u to $d(v) + w(v,u)$.

return d .

Time complexity: $O(n)$ insert $\Rightarrow O(n)$

$O(n)$ extract $\Rightarrow O(n \log n)$

$O(m)$ decrease $\Rightarrow O(m)$

Total: $O(n+m+n \log n)$
 $= O(m+n \log n)$

Lemma: After each iteration, $d(v) = \text{dist}(s,v)$ for all $v \in S$.

Proof: Base case: $S = \{s\}$. $d(s) = 0$ as wanted.

IS: Assume IH for S .

Let $v \in V - S$, $u \in S$ st. $d(v) = d(u) + w(u,v)$ is minimum from algorithm

Case 1: If v is not reachable from S , then $\text{dist}(s,v) = \infty$.

Case 2: If v is reachable from S , let $p_v: s \rightsquigarrow v$ be shortest.

$\therefore s \in S$, $v \notin S$

$\therefore \exists u' \rightarrow x$ in p_v st. $u' \in S$, $x \notin S$

$\therefore p_v: s \rightsquigarrow u' \rightarrow x \rightsquigarrow v$
_{path edge p'}

$\therefore \text{dist}(s,v) = \underbrace{d(u') + w(u',x)}_{\text{IH}} + w(x,v) \geq d(u') + w(u',x) \geq d(u') + w(u',v) = d(v)$
_{nonnegative weight}

$\therefore d(v) = \text{dist}(s,v)$

Efficient:

Instead of setting S , maintain a priority-queue Q for $V - S$.

Key: each $v \in V - S$, priority: $d_v = \min_u (d(u) + w(u,v))$.

Operations: Amortized

Insert v into PQ with priority d $O(1)$

Decrease the priority of v to d if lower $O(1)$

Extract the key v with min priority $O(\log n)$